TAEKJUN (TJ) LEE

SOFTWARE ENGINEER | BUSINESS SYSTEMS AND AUTOMATION

Seoul, Republic of Korea

leetj94@gmail.com | LinkedIn | Portfolio and case studies | Employer sponsorship required

PROFILE

Software engineer at Samsung Electronics, building internal business systems since 2020. Turns operational workflows into Python/FastAPI services, Vue.js applications and integrated data pipelines. Owns requirements, architecture, implementation, automated testing, deployment and production support. Delivered cross-system automation that reduced registration work by 87%, while helping users adopt and operate new systems.

PROFESSIONAL EXPERIENCE

Samsung Electronics | Device Solutions Jan 2019 - Present

Systems Engineer 2020 - Present

- Own business applications from requirements and architecture through database design, frontend/backend implementation, AWS/on-premises deployment and ongoing support.
- Collaborate through Git branches, pull requests and code reviews; write automated tests and maintain CI/CD, production deployments, monitoring and incident response.
- Work directly with engineering users and vendors to define requirements, evaluate solutions and improve products through user feedback.

Facilities Engineer 2019 - 2020

- Operated infrastructure equipment and investigated anomalies using quality and maintenance data, developing firsthand understanding of users' workflows and operational constraints.

SELECTED SYSTEMS AND AUTOMATION PROJECTS

Engineering Document Delivery Platform 2025 - 2026

- Replaced email-based contractor handoffs with a web workflow for submission, automated extraction and validation, correction feedback and final employee approval.
- Designed database structures and REST APIs; built FastAPI services, Vue.js screens and AWS integrations using S3, EventBridge, SQS, Lambda and RDS, with Autodesk APS document processing.
- Implemented retries, duplicate detection and idempotent ingestion, with role-based access and audit logging.
- Configured EC2 deployment and CodePipeline-based CI/CD with automated tests, health checks and rollback procedures.
- Platform scope: approximately 30,000-70,000 drawings and 10,000 registered assets/components per FAB.

Facility Master Data Platform 2020 - 2021

- Defined common equipment identifiers, data standards and lifecycle workflows across quality, maintenance, operations and work systems with different registration requirements.
- Prepared approximately **one million rows** across 50 columns; built Python/SQL validation, ETL/EAI integrations and Spotfire reconciliation reports across system databases.
- Reduced engineer time for equivalent cross-system registration work by **87%**; measured cross-system discrepancy rates fell from **17% to 0%**.
- Handled production requests, incidents and user training for approximately one year; extended features and integrations to additional business systems.

ADDITIONAL SYSTEMS AND AUTOMATION PROJECTS

3D Digital Twin Platform 2023 - 2024; ongoing

- Built application and integration layers connecting **eight operational systems** in a searchable workspace for approximately 1,000 users and **1.5 million** assets/components.
- Linked equipment location with quality trends, maintenance history and work status. Evaluated NVIDIA, Siemens and AVEVA technologies through vendor discussions and proofs of concept.
- Developed Vue interfaces, Python services and ETL/EAI integrations; delivered on-premises through Git, Jenkins and Kubernetes-based CI/CD.

Drawing Management and Search 2021 - 2022

- Redesigned drawing handover processes and standardised equipment-to-drawing relationships so users could find relevant drawings by equipment rather than opening files individually.
- Designed extraction and integration workflows and automated completeness/design-rule checks for a platform serving approximately **4,500 users**.

Data Governance and Operational Automation 2023 - 2026

- Managed data from eight infrastructure systems for a data lake initiative; defined metadata, security classifications and disclosure/access rules.
- Built a FastAPI/Oracle/Vue.js ETL operations dashboard with job-level logs, failure alerts and retry controls to support troubleshooting and recovery.
- Authored reusable team guidance for API standards, UI design, testability and fallback behaviour. Taught SQL and Spotfire within an engineering organisation of approximately 2,000 employees.
- Contributed to AI-ready data eligibility work in 2026 and prototyped workflow/data integrations using external language models.

TECHNICAL SKILLS

Development: Python, SQL, JavaScript, FastAPI, Vue.js, REST APIs, Pydantic, pytest

Data and integration: Data modelling and quality, ETL/EAI, Oracle, PostgreSQL, MySQL/MariaDB, SQL Server, Spotfire

Cloud and delivery: AWS EC2, S3, Lambda, SQS, EventBridge, RDS; Git, Jenkins, CodePipeline, CI/CD, Kubernetes, Linux, Nginx

Business systems: Workflow analysis, data standards, validation rules, user training, MES/MIS, Autodesk APS/ACC, Revit/BIM

EDUCATION AND LANGUAGES

M.S. Industrial Engineering | Sungkyunkwan University Feb 2026

GPA 4.38/4.5

B.S. Mechanical Engineering | Sungkyunkwan University Feb 2019

Korean: native | English: Fluent

RECOGNITION

Samsung Best Contributor/Performer (2022, 2023). Project Incentive Awards: Master Data (2021), Drawings (2022), Digital Twin (2023), 3D Integration (2024).